# EHM(X): 制約生成と solver の合成

型付け・算法・健全性・主要性・条件付き完全性

## 1 構文とパラメータ

$$
\begin{aligned}
&e ::= i \mid n \mid b \mid \lambda.e \mid e\,e \\
&\quad \mid \mathbf{let}\ e\ \mathbf{in}\ e \mid \mathbf{perform}\ o\ e \mid \mathbf{handle}_o(e, e_r, e_o), \\
&v ::= i \mid n \mid b \mid \lambda.e, \qquad i, o \in \mathbb{N}, \\
&\epsilon = (L, M) \in \mathcal{P}_f(\mathbb{N})^2, \quad \langle\epsilon\rangle_\rho = L \cup \bigcup_{\mu \in M} \rho_E(\mu), \\
&A = (A_T, A_E) \in \mathcal{P}_f(\mathbb{N})^2, \\
&\rho = (\rho_T : \mathbb{N} \to V, \rho_E : \mathbb{N} \to \mathcal{P}_f(\mathbb{N})).
\end{aligned}
$$

項変数は de Bruijn 添字. 操作節：$0 = k,\ 1 = x$.

$$
\begin{gathered}
T : \text{型式}, \quad V \neq \varnothing, \quad \alpha_i,\ \mathrm{Int},\ \mathrm{Bool},\ t \xrightarrow{\epsilon} u \in T, \quad \leq \subseteq V^2, \\
a \leq a, \qquad (a \leq b \wedge b \leq c) \Rightarrow a \leq c, \qquad (a \leq b \wedge b \leq a) \Rightarrow a = b, \\
a' \leq a \wedge b \leq b' \Rightarrow a \xrightarrow{E} b \leq a' \xrightarrow{E} b'.
\end{gathered}
$$

$T$ の構成子の網羅性・単射性は仮定しない.

$$
\begin{aligned}
\langle\alpha_i\rangle_\rho &= \rho_T(i), & \langle\mathrm{lit}(b)\rangle_\rho &= \mathrm{groundLit}(b), \\
\langle a \xrightarrow{\epsilon} b\rangle_\rho &= \mathrm{groundArrow}(\langle a\rangle_\rho, \langle\epsilon\rangle_\rho, \langle b\rangle_\rho), & & \\
\langle\mathrm{rename}(f, g, t)\rangle_\rho &= \langle t\rangle_{(\rho_T \circ f, \rho_E \circ g)}, & \theta\alpha_i &= \theta_T(i).
\end{aligned}
$$

$$
\frac{\forall i.\ A \cap \mathrm{supp}(\theta_T(i)) = \varnothing \quad \forall j.\ A_E \cap \mathrm{FV}_E(\theta_E(j)) = \varnothing}{A \cap \mathrm{supp}(\theta t) = \varnothing}
$$

$$
\begin{aligned}
\theta^*\rho &= (i \mapsto \langle\theta_T(i)\rangle_\rho, j \mapsto \langle\theta_E(j)\rangle_\rho), & \mathrm{dom}\,\theta &: \text{有限}, \\
\langle\theta t\rangle_\rho &= \langle t\rangle_{\theta^*\rho}, & \rho \models \theta C &\iff \theta^*\rho \models C, \\
\rho =_{\mathrm{supp}\,t} \nu &\Rightarrow \langle t\rangle_\rho = \langle t\rangle_\nu, & \rho =_{\mathrm{supp}\,C} \nu &\Rightarrow (\rho \models C \iff \nu \models C), \\
\theta|_{\mathrm{supp}\,t} = \mathrm{id} &\Rightarrow \theta t = t, & \mathrm{supp}(\alpha_i) &= (\{i\}, \varnothing).
\end{aligned}
$$

$$
\begin{aligned}
\rho \models \exists A.C &\iff \exists \nu.\ \nu =_{\overline{A}} \rho \wedge \nu \models C, \\
\rho \models \forall B.\exists J.C &\iff \forall \nu =_{\overline{B}} \rho.\ \exists \omega =_{\overline{J}} \nu.\ \omega \models C, \\
C \models_X D &\iff \forall \rho.\ (\rho \models C \Rightarrow \rho \models D), \\
C \equiv_X D &\iff C \models_X D \wedge D \models_X C, & \mathrm{Sat}(C) &\iff \exists \rho.\ \rho \models C, \\
i \in \mathrm{FV}_T(C) &\iff \exists \{i\}_T.C \not\equiv_X C, & \mathrm{FV}(C) &\subseteq \mathrm{supp}\,C.
\end{aligned}
$$

$C, D \in \mathcal{C}_X$. $\top, \wedge, =, \leq, \subseteq, o \notin \epsilon$ は各意味値で解釈する.

## 2 宣言的型付け

$$\sigma ::= t \mid \forall A.D \Rightarrow t, \qquad \Gamma ::= [] \mid \sigma :: \Gamma, \qquad \mathrm{supp}(\forall A.D \Rightarrow t) = (\mathrm{supp}\, D \cup \mathrm{supp}\, t) \setminus A.$$

$C$ が共通の前提では省略する．

$$\frac{\Gamma[i] = \sigma}{\Gamma \vdash i : \sigma!\varnothing} \qquad \frac{}{\Gamma \vdash n : \mathrm{Int}!\varnothing} \qquad \frac{}{\Gamma \vdash b : \mathrm{Bool}!\varnothing}$$

$$\frac{a :: \Gamma \vdash e : b!\eta}{\Gamma \vdash \lambda.e : (a \xrightarrow{\eta} b)!\varnothing} \qquad \frac{\Gamma \vdash f : (a \xrightarrow{\eta} b)!\epsilon_f \quad \Gamma \vdash x : a!\epsilon_x}{\Gamma \vdash f\, x : b!(\epsilon_f \cup \epsilon_x \cup \eta)}$$

$$\frac{\mathrm{Value}(v) \quad \Gamma \vdash v : \sigma!\varnothing \quad \sigma :: \Gamma \vdash e : t!\epsilon}{\Gamma \vdash \mathbf{let}\ v\ \mathbf{in}\ e : t!\epsilon}$$

$$\frac{\neg\, \mathrm{Value}(u) \quad \Gamma \vdash u : a!\eta \quad a :: \Gamma \vdash e : t!\epsilon}{\Gamma \vdash \mathbf{let}\ u\ \mathbf{in}\ e : t!(\eta \cup \epsilon)}$$

$$\frac{C; \Gamma \vdash e : a!\epsilon \quad C \models_X a \le b}{C; \Gamma \vdash e : b!\epsilon} \qquad \frac{C; \Gamma \vdash e : \sigma!\epsilon \quad C \models_X \epsilon = \eta}{C; \Gamma \vdash e : \sigma!\eta}$$

$$\frac{\mathrm{Value}(v) \quad A \cap \mathrm{FV}(C, \Gamma) = \varnothing \quad C \wedge D; \Gamma \vdash v : t!\varnothing}{C \wedge \exists A.D; \Gamma \vdash v : (\forall A.D \Rightarrow t)!\varnothing}$$

$$\frac{C; \Gamma \vdash e : (\forall A.D \Rightarrow t)!\epsilon \quad \mathrm{dom}\,\theta \subseteq A \quad C \models_X \theta D}{C; \Gamma \vdash e : \theta t!\epsilon}$$

$$\frac{C; \Gamma \vdash e : \sigma!\epsilon \quad A \cap \mathrm{FV}(\Gamma, \sigma, \epsilon) = \varnothing}{\exists A.C; \Gamma \vdash e : \sigma!\epsilon} \qquad \frac{C; \Gamma \vdash e : \sigma!\epsilon \quad D \models_X C}{D; \Gamma \vdash e : \sigma!\epsilon}$$

$$\frac{C; \Gamma \vdash e : \sigma!\epsilon \quad \sigma \equiv_\alpha \sigma'}{C; \Gamma \vdash e : \sigma'!\epsilon}$$

$$\begin{aligned}
\mathrm{Adm}_S(C) &\iff \exists R.\ S(R) \wedge C \equiv_X R, \\
\mathrm{Adm}_S(t) &\iff \top, \qquad\qquad \mathrm{Adm}_S(\forall A.D \Rightarrow t) \iff \mathrm{Adm}_S(D), \\
\mathrm{Adm}_S(\Gamma) &\iff \forall \sigma \in \Gamma.\ \mathrm{Adm}_S(\sigma), \\
\mathrm{Typing}_S(C, \Gamma, e, \sigma, \epsilon) &\iff \exists \mathcal{D} : (C; \Gamma \vdash e : \sigma!\epsilon).\ \mathrm{Valid}_S(\mathcal{D}).
\end{aligned}$$

$\mathrm{Valid}_S$ は導出の全節点を検査する．以下 $S = \mathrm{Sat}$.

$$\frac{\mathcal{D} : (C; \Gamma \vdash e : \sigma!\epsilon) \quad \mathrm{Sat}(C) \quad \mathrm{Adm}_{\mathrm{Sat}}(\Gamma)}{\mathrm{Valid}_{\mathrm{Sat}}(\mathcal{D})}$$

# 3 操作・handler・環境

$$\Sigma(o) = (B, a, b), \qquad \mathrm{Closed}(\Sigma) \iff \forall o, (B, a, b) = \Sigma(o).\ \mathrm{supp}(a, b) \subseteq B.$$

$$\frac{\Sigma(o) = (B, a, b) \quad \mathrm{Closed}(B, a, b) \quad \mathrm{dom}\,\theta \subseteq B \quad C; \Gamma \vdash u : \theta a ! \eta}{C; \Gamma \vdash \mathbf{perform}\ o\ u : \theta b ! (\{o\} \cup \eta)}$$

$$\mathrm{Split}_o(\eta, \delta) = \delta \subseteq \eta \wedge \eta \subseteq \{o\} \cup \delta \wedge o \notin \delta.$$

操作宣言の新鮮な改名を $(B, a, b)$ とする.

$$\frac{\begin{array}{c} C; \Gamma \vdash e : t_i ! \epsilon_i \qquad C; t_i :: \Gamma \vdash e_r : t ! \epsilon_r \\ D; (b \xrightarrow{\chi} t) :: a :: \Gamma \vdash e_o : t_o ! \epsilon_o \\ C \models_X \forall B. \exists J. (D \wedge t_o \leq t \wedge \epsilon_o \subseteq \chi) \\ C \models_X \mathrm{Split}_o(\epsilon_i, \delta) \wedge \delta \subseteq \chi \wedge \epsilon_r \subseteq \chi \\ \mathrm{Scope}(B, J; C, \Gamma, t_i, t, a, b, \epsilon_i, \epsilon_r, \delta, \chi) \end{array}}{C; \Gamma \vdash \mathbf{handle}_o(e, e_r, e_o) : t ! \chi}$$

$$\mathrm{Scope} \iff \begin{cases} B \cap \mathrm{FV}(\Gamma) = \varnothing, \\ B \cap \mathrm{supp}(t_i, t) = \varnothing, \\ B_E \cap \mathrm{FV}_E(\epsilon_i, \epsilon_r, \delta, \chi) = \varnothing, \\ J \cap \big(\mathrm{supp}\,\Gamma \cup \mathrm{supp}(t_i, t, a, b) \cup B\big) = \varnothing, \\ J \cap \mathrm{FV}(C) = \varnothing, \\ J_E \cap \mathrm{FV}_E(\epsilon_i, \epsilon_r, \delta, \chi) = \varnothing. \end{cases}$$

$$\frac{C \models_X a \leq b}{C \vdash a \preceq b} \qquad \frac{\mathrm{dom}\,\theta \subseteq A \quad C \models_X \theta D \quad C \vdash \theta t \preceq \sigma}{C \vdash (\forall A. D \Rightarrow t) \preceq \sigma}$$

$$\frac{A \cap \mathrm{FV}(C, \sigma) = \varnothing \quad C \wedge D \vdash \sigma \preceq t}{C \wedge \exists A. D \vdash \sigma \preceq (\forall A. D \Rightarrow t)} \qquad \frac{C \equiv_X D \quad C \vdash \sigma \preceq \tau}{D \vdash \sigma \preceq \tau}$$

$\preceq$ は左右の $\alpha$ 同値で閉じる.

$$\begin{aligned} \mathrm{Inst}_\rho(t) &= \{u \mid \langle t \rangle_\rho \leq u\}, \\ \mathrm{Inst}_\rho(\forall A. D \Rightarrow t) &= \{u \mid \exists \nu =_{\overline{A}} \rho.\ \nu \models D \wedge \langle t \rangle_\nu \leq u\}, \\ \lfloor \Gamma \rfloor_\rho &= \mathrm{map}(\mathrm{Inst}_\rho, \Gamma), \\ \mathrm{GReal}(\Gamma, \rho) &\iff \forall \sigma \in \Gamma.\ \mathrm{Inst}_\rho(\sigma) \neq \varnothing, \\ \mathrm{Real}_C(\Gamma) &\iff \forall \sigma \in \Gamma.\ \exists t.\ C \vdash \sigma \preceq t. \end{aligned}$$

$$\frac{\mathrm{Real}_C(\Gamma) \quad \rho \models C}{\mathrm{GReal}(\Gamma, \rho)} \qquad \frac{C; \Gamma \vdash e : t ! \epsilon \quad \rho \models C \quad \mathrm{GReal}(\Gamma, \rho)}{\lfloor \Gamma \rfloor_\rho \vdash_g e : \langle t \rangle_\rho ! \langle \epsilon \rangle_\rho}$$

$\Gamma' = \theta\Gamma$ （意味的環境代入） $\iff$ $\forall \rho.\ \lfloor \Gamma' \rfloor_\rho = \lfloor \Gamma \rfloor_{\theta^* \rho}$.

# 4　固定した制約生成算法

$N$ は簡約器：$N(D) \equiv_X D$．$\mathcal{A}(e, \Gamma, t, \epsilon; F)$ は部分関数．

$$
\begin{aligned}
f_T^F(i) &= 1 + \max(F_T \cup \{0\}) + i, & f_E^F(i) &= 1 + \max(F_E \cup \{0\}) + i,\\
P(\Gamma, t, \epsilon) &= \operatorname{supp}\Gamma \cup \operatorname{supp} t \cup (\varnothing, \mathrm{FV}_E\,\epsilon),\\
\mathrm{Check}(D, a, t, \eta, \epsilon) &= \top \wedge D \wedge (a \le t) \wedge (\eta = \epsilon),\\
\mathrm{gen}_\Gamma(D, a) &= \forall A.D \Rightarrow a, & A &= (\operatorname{supp} D \cup \operatorname{supp} a) \setminus \operatorname{supp}\Gamma.
\end{aligned}
$$

各再帰呼出しが成功したとき，以下の制約を返す．未定義な変数・操作では missing.

$$
\begin{aligned}
\mathcal{A}(i, \Gamma, t, \epsilon; F) &= \begin{cases} \mathrm{Check}(\top, a, t, \varnothing, \epsilon) & \Gamma[i] = a,\\ \mathrm{Check}(\theta D, \theta a, t, \varnothing, \epsilon) & \Gamma[i] = \forall A.D \Rightarrow a, \end{cases}\\
\theta &= [A \mapsto f^{F \cup \operatorname{supp}\top}(A)],\\
\mathcal{A}(n, \Gamma, t, \epsilon; F) &= \mathrm{Check}(\top, \mathrm{Int}, t, \varnothing, \epsilon),\\
\mathcal{A}(b, \Gamma, t, \epsilon; F) &= \mathrm{Check}(\top, \mathrm{Bool}, t, \varnothing, \epsilon).
\end{aligned}
$$

$$
\begin{gathered}
a = \alpha_{f_T^F(0)}, \quad b = \alpha_{f_T^F(1)}, \quad \eta = \mu_{f_E^F(0)},\\
F' = F \cup P(a :: \Gamma, b, \eta) \cup (\{f_T^F(0), f_T^F(1)\}, \{f_E^F(0)\}),\\
D = \mathcal{A}(e, a :: \Gamma, b, \eta; F'),\\
\boxed{\mathcal{A}(\lambda.e, \Gamma, t, \epsilon; F) = \mathrm{Check}(D, a \xrightarrow{\eta} b, t, \varnothing, \epsilon)}.
\end{gathered}
$$

$$
\begin{gathered}
a = \alpha_{f_T^F(0)}, \quad b = \alpha_{f_T^F(1)}, \quad (\eta_f, \eta_x, \eta) = (\mu_{f_E^F(0)}, \mu_{f_E^F(1)}, \mu_{f_E^F(2)}),\\
K = F \cup P(\Gamma, a, \eta_f) \cup P(\Gamma, b, \eta_x) \cup (\varnothing, \mathrm{FV}_E\,\eta),\\
D_f = \mathcal{A}(f, \Gamma, a, \eta_f; K), \quad D_x = \mathcal{A}(x, \Gamma, b, \eta_x; K \cup \operatorname{supp} D_f),\\
\boxed{\mathcal{A}(f\,x, \Gamma, t, \epsilon; F) = \top \wedge D_f \wedge D_x \wedge a \le b \xrightarrow{\eta} t \wedge \eta_f \cup \eta_x \cup \eta = \epsilon}.
\end{gathered}
$$

$$
\begin{gathered}
\mathrm{Value}(v), \quad a = \alpha_{f_T^F(0)}, \quad D_v = \mathcal{A}(v, \Gamma, a, \varnothing; F \cup P(\Gamma, a, \varnothing)),\\
\sigma = \mathrm{gen}_\Gamma(N(D_v), a), \quad A = (\operatorname{supp} N(D_v) \cup \operatorname{supp} a) \setminus \operatorname{supp}\Gamma, \quad Q = \exists A.N(D_v),\\
D_b = \mathcal{A}(e, \sigma :: \Gamma, t, \epsilon; F \cup P(\sigma :: \Gamma, t, \epsilon) \cup \operatorname{supp} Q),\\
\boxed{\mathcal{A}(\mathbf{let}\ v\ \mathbf{in}\ e, \Gamma, t, \epsilon; F) = \top \wedge Q \wedge D_b}.
\end{gathered}
$$

$$
\begin{gathered}
\neg\,\mathrm{Value}(v), \quad a = \alpha_{f_T^F(0)}, \quad (\eta_v, \eta_b) = (\mu_{f_E^F(0)}, \mu_{f_E^F(1)}),\\
K = F \cup P(\Gamma, a, \eta_v) \cup P(a :: \Gamma, t, \eta_b),\\
D_v = \mathcal{A}(v, \Gamma, a, \eta_v; K), \quad D_b = \mathcal{A}(e, a :: \Gamma, t, \eta_b; K \cup \operatorname{supp} D_v),\\
\boxed{\mathcal{A}(\mathbf{let}\ v\ \mathbf{in}\ e, \Gamma, t, \epsilon; F) = \top \wedge D_v \wedge D_b \wedge (\eta_v \cup \eta_b = \epsilon)}.
\end{gathered}
$$

# 5　操作の生成と射影

$$\Sigma(o) = (B, a, b), \quad \theta = [B \mapsto f^F(B)], \quad K = F \cup \mathrm{supp}(\theta a, \theta b), \quad \eta = \mu_{f_E^K(0)},$$
$$D = \mathcal{A}(u, \Gamma, \theta a, \eta; K \cup P(\Gamma, \theta a, \eta)),$$
$$\boxed{\mathcal{A}(\mathbf{perform}\ o\ u, \Gamma, t, \epsilon; F) = \mathrm{Check}(D \wedge \theta a \le \theta a, \theta b, t, \{o\} \cup \eta, \epsilon)}.$$

**Handler**

$$\begin{aligned}
t_i &= \alpha_{f_T^F(0)}, & t_r &= \alpha_{f_T^F(1)},\\
(\epsilon_i, \epsilon_r, \delta) &= (\mu_{f_E^F(0)}, \mu_{f_E^F(1)}, \mu_{f_E^F(2)}),\\
K &= F \cup \mathrm{supp}(t_i, t_r) \cup (\varnothing, \mathrm{FV}_E(\epsilon_i, \epsilon_r, \delta)),\\
D_i &= \mathcal{A}(e, \Gamma, t_i, \epsilon_i; K),\\
K_r &= K \cup \mathrm{supp}\, D_i, & D_r &= \mathcal{A}(e_r, t_i :: \Gamma, t_r, \epsilon_r; K_r),\\
K_o &= K_r \cup \mathrm{supp}\, D_r, & (B, a, b) &= \mathrm{rename}_{f^{K_o}}(\Sigma(o)),\\
\Gamma_o &= (b \xrightarrow{\chi} t_r) :: a :: \Gamma,\\
F_o &= K_o \cup P(\Gamma_o, t_r, \chi) \cup \mathrm{supp}(a, b) \cup B,\\
\epsilon_o &= \mu_{f_E^{F_o}(0)}, & D_o &= \mathcal{A}(e_o, \Gamma_o, t_r, \epsilon_o; F_o \cup P(\Gamma_o, t_r, \epsilon_o)),\\
J &= (\mathrm{supp}\, D_o \cup (\varnothing, \mathrm{FV}_E\, \epsilon_o)) \setminus F_o.
\end{aligned}$$

$$\begin{aligned}
H = {} & \top \wedge D_i \wedge D_r \wedge t_r \le t_r\\
& \wedge \big(\forall B. \exists J. (D_o \wedge t_r \le t_r \wedge \epsilon_o \subseteq \chi)\big)\\
& \wedge \mathrm{Split}_o(\epsilon_i, \delta) \wedge \delta \subseteq \chi \wedge \epsilon_r \subseteq \chi,\\
\mathcal{A}(\mathbf{handle}_o(e, e_r, e_o), \Gamma, t, \chi; F) = {} & \mathrm{Check}(H, t_r, t, \chi, \chi).
\end{aligned}$$

**外側の射影と solver 呼出し**

$$\begin{aligned}
F &= P(\Gamma, t, \epsilon) \cup \mathrm{supp}\, \top, \quad D = \mathcal{A}(e, \Gamma, t, \epsilon; F),\\
\mathcal{P}(\Gamma, e, t, \epsilon) &= \exists(\mathrm{supp}\, D \setminus F).D,\\
\mathcal{I}^{\mathrm{at}}_{N,S}(\Gamma, e, t, \epsilon) &= S(\top, \mathcal{P}(\Gamma, e, t, \epsilon)),\\
p &= f_T^{\mathrm{supp}\,\Gamma}(0), \quad q = f_E^{\mathrm{supp}\,\Gamma}(0),\\
\mathcal{I}_{N,S}(\Gamma, e) &= \mathcal{I}^{\mathrm{at}}_{N,S}(\Gamma, e, \alpha_p, \mu_q).
\end{aligned}$$

missing は後続へ伝播する．solved$(R, \theta)$ の結果は

$$(R, \theta\Gamma, \theta\alpha_p, \theta\mu_q).$$

閉環境では $\mathrm{generalize}(\Gamma, R, \theta\alpha_p, \theta\mu_q)$ を返す型スキームとする．

# 6 solver の契約

$$S : \mathcal{C}_X \times \mathcal{C}_X \longrightarrow \mathsf{solved}(R,\theta) \mid \mathsf{unresolved}(R) \mid \mathsf{inconsistent},$$
$$[\theta] = \bigwedge_{i \in \mathrm{dom}_T\,\theta} (\alpha_i = \theta_T(i)) \wedge \bigwedge_{j \in \mathrm{dom}_E\,\theta} (\mu_j = \theta_E(j)),$$
$$\mathrm{SolvedSubst}(\theta) \iff \begin{cases} \forall i \in \mathrm{dom}_T\,\theta.\ \mathrm{dom}\,\theta \cap \mathrm{supp}(\theta_T(i)) = \varnothing, \\ \forall j \in \mathrm{dom}_E\,\theta.\ \mathrm{dom}_E\,\theta \cap \mathrm{FV}_E(\theta_E(j)) = \varnothing. \end{cases}$$

**部分 solver**（S1・S2・P1 型の契約 [2]）

$$S(G,W) = \mathsf{solved}(R,\theta) \Longrightarrow \begin{cases} \mathrm{SolvedSubst}(\theta), \\ \mathrm{dom}\,\theta \cap \mathrm{FV}(G,R) = \varnothing, \\ G \wedge R \models_X \theta W, \\ G \wedge W \models_X R \wedge [\theta], \\ \mathrm{Sat}(G \wedge R). \end{cases}$$

$$S(G,W) = \mathsf{unresolved}(R) \Rightarrow G \wedge R \equiv_X G \wedge W,$$
$$S(G,W) = \mathsf{inconsistent} \Rightarrow \neg\,\mathrm{Sat}(G \wedge W).$$

**完全性の追加契約**

$$\boxed{\mathrm{Complete}(S) \iff \forall G, W.\ \mathrm{Sat}(G \wedge W) \Rightarrow \exists R, \theta.\ S(G,W) = \mathsf{solved}(R,\theta)}.$$

$W$ は handler の量化制約も含む．$R = \top$ は要求しない．

**補題 6.1** (解集合の保存)**.**

$$S(G,W) = \mathsf{solved}(R,\theta) \Longrightarrow G \wedge W \equiv_X G \wedge R \wedge [\theta].$$

証明.

$$\rho \models G \wedge W \Rightarrow \rho \models R \wedge [\theta],$$
$$\rho \models G \wedge R \wedge [\theta] \Rightarrow \theta^*\rho \models W \ \wedge\ \theta^*\rho = \rho \Rightarrow \rho \models W.$$

□

**契約のモデル**

$$S_?(G,W) = \mathsf{unresolved}(W),$$
$$S_{\mathrm{sem}}(G,W) = \begin{cases} \mathsf{solved}(W, \mathrm{id}) & \mathrm{Sat}(G \wedge W), \\ \mathsf{inconsistent} & \neg\,\mathrm{Sat}(G \wedge W). \end{cases}$$

$$\mathrm{PartialSolver}(S_?) \qquad \mathrm{PartialSolver}(S_{\mathrm{sem}}) \wedge \mathrm{Complete}(S_{\mathrm{sem}}).$$

$S_{\mathrm{sem}}$ は非計算的なモデル．実行可能な判定手続きは別の実装義務．HM(X) の制約形式と型項再構成の区別は [1, §4.4].

# 7　生成と合成の定理

この節の健全性では Closed($\Sigma$) を仮定する.

**定理 7.1** (固定した生成算法)**.**

$$\begin{gathered} P(\Gamma,t,\epsilon)\subseteq F,\quad \mathcal{A}(e,\Gamma,t,\epsilon;F)=D \\ \Longrightarrow\quad \top;\Gamma\vdash_N e:t!\epsilon\Rightarrow D, \\ \exists\rho.\ \lfloor\Gamma\rfloor_\rho\vdash_g e:\langle t\rangle_\rho!\langle\epsilon\rangle_\rho \\ \Longrightarrow\quad \exists D.\ \mathcal{A}(e,\Gamma,t,\epsilon;F)=D\ \wedge \\ \forall\rho.\ \lfloor\Gamma\rfloor_\rho\vdash_g e:\langle t\rangle_\rho!\langle\epsilon\rangle_\rho\Rightarrow\exists\nu=_F\rho.\ \nu\models D. \end{gathered}$$

証明. $e$ に関する構造帰納法.

$$\begin{aligned} \lambda,\mathrm{app},\mathrm{let},\mathrm{perform}:\quad & \rho\xmapsto{\mathrm{fresh}(F)}\rho_0\xmapsto{\mathrm{IH}_1}\rho_1\xmapsto{\mathrm{IH}_2}\rho_2, \\ & \rho_{i+1}=_{F\cup\mathrm{supp}\,D_1\cup\cdots\cup\mathrm{supp}\,D_i}\rho_i; \\ \mathrm{handler}:\quad & \forall\rho_B.\ \exists\rho_J.\ \rho_J\models D_o\wedge t_r\le t_r\wedge\epsilon_o\subseteq\chi, \\ & J\cap F_o=\varnothing,\quad B\cap K_o=\varnothing,\quad \mathrm{Scope}(B,J;H,\ldots). \end{aligned}$$

□

**定理 7.2** (射影後の正確性)**.**

$$\mathcal{P}(\Gamma,e,t,\epsilon)=D\wedge\mathrm{GReal}(\Gamma,\rho)\Longrightarrow\big(\rho\models D\iff\lfloor\Gamma\rfloor_\rho\vdash_g e:\langle t\rangle_\rho!\langle\epsilon\rangle_\rho\big).$$

証明. ($\Rightarrow$)：生成健全性と導出の意味解釈. ($\Leftarrow$)：$\nu=_F\rho$ を前定理で得て，$\omega=\rho[\mathrm{supp}\,D_{\mathrm{raw}}\setminus F\mapsto\nu]$ を存在証人とする. □

**定理 7.3** (成功時の健全性)**.**

$$\begin{gathered} \mathcal{I}^{\mathrm{at}}_{N,S}(\Gamma,e,t,\epsilon)=\mathsf{solved}(R,\theta),\quad \rho\models R,\quad \mathrm{GReal}(\Gamma,\theta^*\rho) \\ \Longrightarrow\quad \lfloor\Gamma\rfloor_{\theta^*\rho}\vdash_g e:\langle\theta t\rangle_\rho!\langle\theta\epsilon\rangle_\rho. \end{gathered}$$

$$\Gamma'=\theta\Gamma,\quad \mathrm{Real}_\top(\Gamma),\quad \mathrm{Adm}_{\mathrm{Sat}}(\Gamma')\quad\Longrightarrow\quad \mathrm{Typing}_{\mathrm{Sat}}(R,\Gamma',e,\theta t,\theta\epsilon).$$

証明. $\rho\models R\Rightarrow\theta^*\rho\models D\Rightarrow\lfloor\Gamma\rfloor_{\theta^*\rho}\vdash_g e:\langle t\rangle_{\theta^*\rho}!\langle\epsilon\rangle_{\theta^*\rho}$. $\mathrm{Sat}(R)$ と `Typing.of_ground` により再構成する. □

**定理 7.4** (solver の契約を仮定した完全性)**.**

$$\boxed{\mathrm{Complete}(S)\wedge\mathrm{Real}_C(\Gamma)\wedge\mathrm{Typing}_{\mathrm{Sat}}(C,\Gamma,e,\sigma,\epsilon)\Rightarrow\exists R,\theta.\ \mathcal{I}_{N,S}(\Gamma,e)=\mathsf{solved}(R,\theta)}.$$

証明. $\rho\models C\Rightarrow\lfloor\Gamma\rfloor_\rho\vdash_g e:u!E$. 新鮮な結果名に $(u,E)$ を割り当てる. 生成完全性より $\mathrm{Sat}(D)$, $\mathrm{Complete}(S)$ より $S(\top,D)=\mathsf{solved}(R,\theta)$. □

# 8 実際の出力の主要性

$$\frac{A \cap \mathrm{FV}(\Gamma) = \varnothing \quad \mathrm{Adm}_{\mathrm{Sat}}(\exists A.C) \quad \mathrm{Typing}_{\mathrm{Sat}}(C, \Gamma, e, t, \epsilon)}{\mathrm{ComputationTyping}_{\mathrm{Sat}}(\exists A.C, \Gamma, e, \forall A.C \Rightarrow (t!\epsilon))}$$

ComputationTyping は許容される制約への強化で閉じる．

$$\begin{aligned}
\mathrm{generalize}(\Gamma, R, t, \epsilon) &= \forall A.R \Rightarrow (t!\epsilon), \\
A &= (\mathrm{supp}\, R \cup \mathrm{supp}\, t \cup (\varnothing, \mathrm{FV}_E\, \epsilon)) \setminus \mathrm{supp}\, \Gamma, \\
s &= \forall A.R \Rightarrow (t!\epsilon), \\
\mathrm{FinInst}(s; C, u, \eta) &\iff \exists \phi(\text{有限}).\ \mathrm{dom}\, \phi \subseteq A \wedge \\
&\qquad \forall \rho \models C.\ \phi^* \rho \models R \wedge \langle t \rangle_{\phi^* \rho} \leq \langle u \rangle_\rho \wedge \langle \epsilon \rangle_{\phi^* \rho} = \langle \eta \rangle_\rho.
\end{aligned}$$

**定理 8.1** (開環境の有限因子)**.**

$$\begin{gathered}
\mathcal{I}^{\mathrm{at}}_{N,S}(\Gamma, e, t, \epsilon) = \mathsf{solved}(R, \theta), \qquad \Gamma' = \sigma\Gamma, \quad \mathrm{Real}_C(\Gamma'), \\
\mathrm{Typing}_{\mathrm{Sat}}(C, \Gamma', e, \sigma t, \sigma\epsilon) \\
\Longrightarrow \quad \exists \phi(\text{有限}).\ \forall \rho \models C.\ \phi^* \rho \models R \wedge \theta^*(\phi^* \rho) = \sigma^* \rho.
\end{gathered}$$

証明. $\phi = \sigma$ とする．

$$\rho \models C \Rightarrow \sigma^* \rho \models D \Rightarrow \sigma^* \rho \models R \wedge [\theta] \Rightarrow \theta^*(\sigma^* \rho) = \sigma^* \rho.$$

□

**定理 8.2** (閉環境の主要計算型・主要値型)**.**

$$\begin{gathered}
\mathrm{Closed}(\Sigma), \quad \mathrm{supp}\, \Gamma = \varnothing, \quad \mathrm{Real}_\top(\Gamma), \quad \mathrm{Adm}_{\mathrm{Sat}}(\Gamma), \\
\mathcal{I}_{N,S}(\Gamma, e) = \mathsf{solved}(R, \theta), \quad p = f_T^{\mathrm{supp}\, \Gamma}(0), \quad q = f_E^{\mathrm{supp}\, \Gamma}(0), \\
s = \mathrm{generalize}(\Gamma, R, \theta\alpha_p, \theta\mu_q) \\
\Longrightarrow \quad \mathrm{ComputationTyping}_{\mathrm{Sat}}(\top, \Gamma, e, s) \wedge \\
\forall C, t, \epsilon.\ \mathrm{Typing}_{\mathrm{Sat}}(C, \Gamma, e, t, \epsilon) \Rightarrow \mathrm{FinInst}(s; C, t, \epsilon).
\end{gathered}$$

$s = \forall A.R \Rightarrow (u!\eta)$ と書く．値なら $(\top, \forall A.R \Rightarrow u)$ は主要対で，即時効果は空．

証明. 生成制約を $D$ とする．$\mathrm{Sat}(R) \wedge R \models \theta D \Rightarrow \mathrm{Sat}(D)$．生成の主要性と実行結果の一意性から，実際の $D$ の有限被覆を得る．

$$\begin{aligned}
\rho \models C \Rightarrow \sigma^* \rho \models D \wedge \langle \alpha_p \rangle_{\sigma^* \rho} \leq \langle t \rangle_\rho \wedge \langle \mu_q \rangle_{\sigma^* \rho} = \langle \epsilon \rangle_\rho, \\
D \equiv_X R \wedge [\theta] \Rightarrow \sigma^* \rho \models R \wedge \theta^*(\sigma^* \rho) = \sigma^* \rho.
\end{aligned}$$

$\sigma$ を $s$ の束縛名へ制限し，同じ有限因子を全ての $\rho \models C$ に用いる．値の場合は `ComputationTyping.principal_value` を適用する． □

$$\begin{aligned}
\mathcal{I}^{\mathrm{at}}_{N,S}(\Gamma, e, t, \epsilon) = \mathsf{inconsistent} &\Rightarrow \neg \exists \rho.\ \lfloor \Gamma \rfloor_\rho \vdash_g e : \langle t \rangle_\rho ! \langle \epsilon \rangle_\rho, \\
\mathcal{I}^{\mathrm{at}}_{N,S}(\Gamma, e, t, \epsilon) = \mathsf{unresolved}(R) &\Rightarrow \exists D.\ \mathcal{P}(\Gamma, e, t, \epsilon) = D \wedge R \equiv_X D.
\end{aligned}$$

# 9　具体化：正則な型クラス制約

$$K = \{k_1, \dots, k_n\}, \quad \mathrm{arity} : K \to \mathbb{N}, \qquad \mathcal{C} = \{C_1, \dots, C_m\},$$
$$t ::= \alpha \mid b \mid t \xrightarrow{\epsilon} t \mid k(t_1, \dots, t_{\mathrm{arity}(k)}), \qquad V = \{t \mid \mathrm{FV}(t) = \varnothing\}, \quad \leq = =,$$
$$D ::= \top \mid \bot \mid D \wedge D \mid t = u \mid C(t) \mid \epsilon \subseteq \eta \mid o \notin \epsilon \mid \exists A.D \mid \forall B.\exists J.D \mid \theta D.$$

$\theta D$ は遅延代入：$\rho \models \theta D \iff \theta^* \rho \models D$.

$$\mathcal{I}_b \subseteq \mathcal{C} \times \mathrm{Base}, \qquad \mathcal{I}(C,k) \in \{\bot\} \cup \prod_{j < \mathrm{arity}(k)} \mathcal{P}(\mathcal{C}),$$
$$\mathcal{I}(C,k) = (P_j)_j \quad \equiv \quad \frac{\bigwedge_j \bigwedge_{D \in P_j} D(\alpha_j)}{C(k(\vec{\alpha}))}.$$

同一 head は一つ．前提は head の真部分型のみ．矢印型のインスタンスは置かない．

$$\mathrm{covers}(P,Q) \iff Q \subseteq \bigcup_{C \in P} (\{C\} \cup \mathrm{sup}(C)),$$
$$\mathrm{WF}(\mathcal{I}) \iff \big(D \in \mathrm{sup}(C) \Rightarrow D < C \wedge \mathrm{sup}(D) \subseteq \mathrm{sup}(C)\big) \wedge$$
$$\big((C,b) \in \mathcal{I}_b \Rightarrow \forall D \in \mathrm{sup}(C).\ (D,b) \in \mathcal{I}_b\big) \wedge$$
$$\big(\mathcal{I}(C,k) = (P_j)_j \Rightarrow \forall D \in \mathrm{sup}(C).\ \exists (Q_j)_j.\ \mathcal{I}(D,k) = (Q_j)_j \wedge \forall j.\, \mathrm{covers}(P_j, Q_j)\big).$$
$$\mathrm{check}(\Delta) = \mathsf{some}(\mathcal{I}, \pi) \iff \mathrm{NoDuplicateHeads}(\Delta) \wedge \mathrm{WF}(\mathcal{I}), \quad \mathcal{I} = \mathrm{environment}(\Delta).$$

**有限状態による存在判定**

$$q(b) = \{C \mid (C,b) \in \mathcal{I}_b\}, \qquad q(u \xrightarrow{E} v) = \varnothing,$$
$$\delta_k(q_1, \dots, q_r) = \{C \mid \mathcal{I}(C,k) = (P_j)_j \wedge \forall j.\ P_j \subseteq q_j\},$$
$$q(k(\vec{u})) = \delta_k(q(u_1), \dots, q(u_r)),$$
$$R_0 = \{\varnothing\} \cup \{q(b) \mid (C,b) \in \mathcal{I}_b\},$$
$$R_{i+1} = R_i \cup \{\delta_k(\vec{q}) \mid k \in K,\ \vec{q} \in R_i^{\mathrm{arity}(k)}\}.$$

**定理 9.1.**

$$\mathrm{WF}(\mathcal{I}) \Rightarrow (\mathcal{I} \vdash C(u) \iff C \in q(u)),$$
$$R = \mathrm{lfp}(R_i), \qquad q \in R \iff \exists u \in V.\ q(u) = q,$$
$$\mathrm{inhabited}(P) = \mathsf{true} \iff \exists q \in R.\ P \subseteq q \iff \exists u.\ \bigwedge_{C \in P} C \in q(u).$$

証明．構造帰納法．飽和の測度は $2^m - |R_i|$．同一変数の述語は $P$ に集約する． □

# 10　効果判定・存在射影・実行結果

効果原子の連言を $F$，出現する名前・ラベルを $M, L$ とする．

$$
\begin{aligned}
Q_0 &= M \times L, & \rho_Q(\mu) &= \{o \mid (\mu, o) \in Q\},\\
\mathrm{allow}_{\epsilon \subseteq \eta}(Q, \mu, o) &\iff \mu \in \mathrm{FV}_E(\epsilon) \Rightarrow o \in \langle \eta \rangle_{\rho_Q},\\
\mathrm{allow}_{p \notin \epsilon}(Q, \mu, o) &\iff \mu \in \mathrm{FV}_E(\epsilon) \Rightarrow o \neq p,\\
Q_{i+1} &= \{(\mu, o) \in Q_i \mid \forall f \in F.\, \mathrm{allow}_f(Q_i, \mu, o)\},\\
Q_* &= \mathrm{gfp}(Q_i), & \mathrm{effectSat}(F) &= [\rho_{Q_*} \models F].
\end{aligned}
$$

**定理 10.1.**

$$\mathrm{effectSat}(F) = \mathsf{true} \iff \exists \rho_E.\, \rho_E \models F.$$

証明. $\rho \models F \Rightarrow (\rho \cap L) \models F$. $Q_0$ 内の任意のモデルは各 $Q_i$ 以下．固定点では変数由来の包含・不在が成立し，残るリテラル条件はモデルの存在から従う．停止測度は $|Q_i|$. □

**射影**

$$
\begin{gathered}
D \models \alpha = t, \quad \alpha \notin \mathrm{FV}_T(t), \quad \alpha \in A_T, \quad \theta = [\alpha \mapsto t] \quad \Longrightarrow \quad \exists A.D \equiv \exists (A \setminus \{\alpha\}).\theta D,\\
D \models \mu = \epsilon, \quad \mu \notin \mathrm{FV}_E(\epsilon), \quad \mu \in A_E \quad \Longrightarrow \quad \exists A.D \equiv \exists (A \setminus \{\mu\}).[\mu \mapsto \epsilon] D,\\
\mathrm{supp}\, D \subseteq A \quad \Longrightarrow \quad \rho \models \exists A.D \iff \mathrm{Sat}(D),\\
N(\forall B. \exists J. D) = \mathrm{smartClause}(B, \varnothing, \mathrm{project}(J, N(D))),\\
\boxed{N(D) \equiv D}.
\end{gathered}
$$

型・効果の束縛をまたぐ未処理の代入は suspension として残す．

$$\alpha \in \mathrm{FV}_T(t), \quad t \neq \alpha \;\Rightarrow\; |\rho_T(\alpha)| < |\langle t \rangle_\rho| \;\Rightarrow\; \neg\, \mathrm{Sat}(\alpha = t).$$

**具体的 solver $S_{\mathrm{tc}}$**

$$
\begin{gathered}
S_{\mathrm{tc}}(G, W) = \mathsf{solved}(R, \theta) \Rightarrow \mathrm{SolvedSubst}(\theta) \wedge \theta G \equiv G \wedge \theta R \equiv R \wedge \mathrm{Sat}(G \wedge R),\\
G \wedge R \models \theta W, \qquad G \wedge W \models R \wedge [\theta],\\
S_{\mathrm{tc}}(G, W) = \mathsf{unresolved}(R) \Rightarrow G \wedge R \equiv G \wedge W,\\
S_{\mathrm{tc}}(G, W) = \mathsf{inconsistent} \Rightarrow \neg\, \mathrm{Sat}(G \wedge W).
\end{gathered}
$$

$S_{\mathrm{tc}}$ は有限探索・構造再帰・上記の停止測度で計算する．意味的な充足可能性 oracle は用いない．健全性・主要性は前節の定理を $X_{\mathrm{tc}}, N, S_{\mathrm{tc}}$ に適用する（同じ環境・宣言の前提）．

# 11　完全性の証明範囲と具体的検証

$$F ::= \top \mid F \wedge F \mid C(\alpha) \mid \epsilon \subseteq \eta \mid o \notin \epsilon,$$
$$\mathcal{D} ::= F \mid \exists A.\mathcal{D}.$$

simplify は構造的簡約，$N$ は存在射影を含む正規化.

**定理 11.1** (具体的 solver の部分体系での完全性)**.**

$$\mathrm{simplify}(G \wedge N(W)) \in \mathcal{D} \wedge \mathrm{Sat}(G \wedge W) \quad \Longrightarrow \quad \exists R, \theta.\ S_{\mathrm{tc}}(G, W) = \mathsf{solved}(R, \theta).$$

型等式の逐次消去も実装するが，全ての handler-free 生成制約が判定域へ到達する証明は未完了．全体系の Complete($S_{\mathrm{tc}}$)，handler 固有の不完全性は主張しない.

$$\mathrm{Eq(Int)},\ \mathrm{Eq(Bool)},\ \mathrm{Ord(Int)},\quad \mathrm{Ord}(\alpha) \Rightarrow \mathrm{Eq}(\alpha),$$
$$C(\alpha) \Rightarrow C(\mathrm{List}\,\alpha), \qquad C(\alpha) \wedge C(\beta) \Rightarrow C(\mathrm{Pair}(\alpha, \beta)), \quad C \in \{\mathrm{Eq}, \mathrm{Ord}\}.$$

| 入力（$G = \top$） | 実行結果 |
|---|---|
| $\mathrm{Eq(List\,Bool)}$ | solved |
| $\mathrm{Ord(List\,Bool)}$ | inconsistent |
| $\mathrm{Eq}(\alpha)$ | $\mathsf{solved}(\mathrm{Eq}(\alpha), \mathrm{id})$ |
| $\exists\alpha.\,\mathrm{Eq}(\alpha)$ | $\mathsf{solved}(\top, \mathrm{id})$ |
| $\alpha = \mathrm{Int}$ | $\mathsf{solved}(\top, [\alpha \mapsto \mathrm{Int}])$ |
| $\alpha = \mathrm{List}\,\alpha$ | inconsistent |
| $\forall\alpha.\exists\beta.\alpha = \beta$ | solved |
| $\exists\beta.\forall\alpha.\alpha = \beta$ | unresolved |

最後の式は偽であり，未解決は検出能力の制限を表す．量化順序の交換は行わない.

$$\Sigma(o) = \forall\alpha.\alpha \to \alpha, \qquad e_h = \mathbf{handle}_o(\mathbf{perform}\ o\ 0_{\mathrm{int}},\ 0,\ 0\ 1),$$
$$[] \vdash_g e_h : \mathrm{Int}!\varnothing, \qquad \mathcal{I}_{N, S_{\mathrm{tc}}}([], e_h) = \mathsf{solved}(R, \theta).$$

後者と多相 let・未使用の Eq 束縛・perform は実行検証．前者は Lean の型付け証明．**対応表**（名前空間 `EHMX.TypeClasses`）

| 保証 | Lean |
|---|---|
| 宣言検査・クラス意味論 | `Declarations.check_iff`, `holds_iff` |
| 共同存在判定 | `reachable_iff`, `inhabited_iff` |
| 効果の完全な判定 | `propagationSat_iff` |
| 平坦な制約の完全な判定 | `flatSat_iff` |
| 型・効果の存在消去 | `eliminate_bound`, `eliminate_effect` |
| 同値正規化 | `normalize_correct` |
| 具体的 solver の契約 | `solver` |
| 具体的推論の健全性・主要性 | `inferClasses_sound`, `inferClasses_principal` |
| 上記 $\mathcal{D}$ の完全性 | `solver_complete_fragment` |
| 具体的 handler の型付け | `Examples.resumeIdentity_ground` |
| 実行検証 | `Examples.regressions` |

## 12　抽象定理の Lean 対応

**対応表**（名前空間 `EHMX.HMX`）

| 数式・保証 | Lean |
|---|---|
| $\mathcal{A}, \mathcal{P}, \mathcal{I}^{\mathrm{at}}, \mathcal{I}$ | `generate`, `generateProjected`, `inferAt`, `infer` |
| 部分・完全 solver | `PartialSolver`, `PartialSolver.Complete` |
| 契約の非計算的モデル | `PartialSolver.semantic_complete` |
| 固定した生成の健全性 | `generate_sound` |
| 固定した生成の完全性 | `Inference.groundComplete` |
| 射影後の正確性 | `generateProjected_ground_iff` |
| 開環境の成功時健全性 | `inferAt_ground_sound`, `inferAt_sound` |
| 開環境の有限因子 | `infer_principal_open` |
| 実際の結果の主要性 | `infer_principal_closed` |
| 期待型なしの条件付き完全性 | `infer_complete` |
| 不整合・未解決 | `inferAt_inconsistent`, `inferAt_unresolved` |
| handler の具体例 | `AlgorithmExamples.resumeIdentity_ground`, `resumeIdentity_generated`, `resumeIdentity_complete`, `resumeIdentity_deferred` |

## 参考文献


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[2] D. Vytiniotis, S. Peyton Jones, T. Schrijvers, M. Sulzmann. *OutsideIn(X): Modular type inference with local assumptions*. JFP, 2011, §§3.6, 6.4. 原典.